

Kith Foundry -- Scale, Capacity & Revenue Analysis

March 12, 2026 | Prepared by GitHub Copilot (Claude Sonnet 4.6)

Q1: Is the application able to support and scale to 1M customers per day?

No -- not even close. Here is the honest breakdown.

Scale Assessment: 1M Customers/Day

1M customers/day ~= 12 req/sec average, easily 120-600 req/sec at peak (assuming 10:1 peak ratio and multiple API calls per session).

Hard Blockers

Issue	Current State	What 1M/day Needs
Single process	One uvicorn process, no multi-worker	Horizontally scaled workers behind a
DB connection pool	pool_size=5, max_overflow=10 -> max	pool_size=20, max_overflow=40 + PgBo
No caching	Every request hits Postgres cold	Redis for hot-path queries (project
No rate limiting	Zero middleware protecting LLM-calli	Per-user/IP rate limits on /csuite,
In-memory global state	_workers, _fix_state all in-process	Must move to Redis -- loses all stat
Fly sandbox cold start	Up to 60s for IP allocation (20 retr	Pre-warming, pooling strategy, or as
WebSocket per project	Single-process tracker, no distribut	Redis pub/sub or a dedicated WS gate

Compounding Problems at Scale

- * C-Suite = 7 parallel LLM calls per request. At 1% of daily users = 70,000 LLM calls/day from that feature alone.
- * retries=0 on all Inngest jobs. At scale, transient failures become frequent -- silent data loss.
- * PROVIDER_ENCRYPTION_KEY not set = random key per boot. Every restart invalidates all stored API keys.
- * BRIDGE_SECRET not set = random per boot. Every backend restart disconnects all live sandbox sessions.
- * No CDN -- static frontend assets at 1M users will saturate egress.

What Must Change Before 1M/day Is Viable

1. Deploy multiple uvicorn workers (--workers 4+ or Gunicorn + uvicorn workers) behind nginx/Caddy.
2. Add PgBouncer (transaction pooling mode) + set pool_size=20, max_overflow=40, pool_recycle=300.
3. Redis for session state, global in-memory dicts, and query caching on hot reads.
4. Set PROVIDER_ENCRYPTION_KEY and BRIDGE_SECRET as stable secrets in deployment environment.
5. Enable Inngest retries (retries=3 minimum) on all functions.
6. Rate limiting middleware (slowapi or reverse proxy) on LLM-heavy endpoints.
7. CDN (Cloudflare, CloudFront) in front of the frontend build.
8. Fly sandbox pre-warming -- queue-based machine creation instead of blocking the request.
9. Database read replicas for project/idea list queries.

Q2: How many users and requests per minute can we support today?

The Three Binding Constraints

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1. Sync SQLAlchemy in async def handlers -- the worst bottleneck. Every db.query() call blocks the asyncio event loop entirely. Remote Postgres latency ~10-50ms avg; typical request 3-5 DB calls = 30-150ms total event loop blocking.
2. DB connection pool -- pool_size=5 (default), max_overflow=10 = max 15 simultaneous connections. Only 1 is exercised at a time anyway due to event loop blocking.
3. Inngest concurrency ceilings -- Design=3, C-Suite=5, Artifacts=3. All retries=0.

Requests Per Minute -- By Endpoint Type

Endpoint Type	Avg Response Time	Est. Sustained Req/Min
Simple reads (GET /projects, auth ch	30-80ms	750-1,200
Project writes (POST /projects)	50-150ms	400-900
C-Suite POST /csuite/{id}/run	15-60s (LLM async)	5-10 (Inngest limit=5)
Design generation	10-30s per mockup	6-18 (Inngest limit=3-5)
Artifacts generation	10-20s	6-15 (Inngest limit=3-5)
Overall mixed workload	--	~600-900 req/min realistic peak

Daily Active Users

Scenario	Logic	Est. DAU
Heavy users (full LLM workflows)	C-Suite: 5 concurrent x ~3 runs/hr;	~330 full-workflow users/day
Realistic mixed (80% light, 20% heav	~600 req/min steady x 10hr x 60 ÷ 20	~3,000-5,000 DAU before degradation

Capacity Summary

Metric	Current Capacity
Peak requests/min (mixed)	~600-900
Simple reads/min only	~1,000-1,200
Concurrent active users	15-30
Concurrent LLM jobs	11
Sustainable DAU (light)	~3,000-5,000
Sustainable DAU (full workflows)	~300-500

Key insight: Swapping to asyncpg + AsyncSession (async SQLAlchemy) would multiply throughput by 5-10x with no infrastructure changes.

Q3: What should we charge, and what is the ideal pricing model vs. the market?

What Kith Foundry Actually Is

This is not just an app builder. It is a full-stack startup acceleration platform -- a single loop from zero to working prototype:

Idea -> Validation -> C-Suite Advisory -> Brand/UI Design -> Live Code -> Business Docs

* Ideation Engine -- questionnaire -> 3 curated startup ideas with TAM, CAC/LTV, 90-day launch plan, strategic moat

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- * C-Suite Simulation -- 7 parallel AI advisors (CEO, CTO, CFO, CMO, CPO, COO, CDO) with financial modeling, go-to-market, tech architecture
- * Design Studio -- brand-consistent UI mockup generation with locked design tokens
- * Live Code Sandbox -- React/TypeScript app with live preview, per-project Fly.io VM
- * Artifact Generation -- PRD, Executive Brief, Tech Spec, Design System docs

Competitive Market Landscape

Product	What it does	Price/mo	Gap vs Kith
Lovable	AI app builder (code gen only)	\$20-\$80	No ideation, no advisory, no strateg
Bolt.new	AI app builder	\$20	No ideation, no advisory, no design
v0 (Vercel)	UI component generation	\$20-\$30	UI only, no business layer
Cursor	AI code editor	\$20	Code only, IDE-based
Tome	AI pitch deck / narrative	\$16-\$25	No code, no advisory
Builder.ai	Managed app building (human+AI)	\$499-\$2,000+	Enterprise, human-assisted, slow

No single product combines all five layers. The closest equivalent bundle = Lovable + v0 + a strategy consultant = \$40-\$80/mo + \$150-\$500/hr consulting.

Recommended Pricing Tiers

Tier	Target	Price	Key Limits
Free	Hobbyists, evaluation	\$0	1 project, 3 C-Suite runs/mo, 5 desi
Indie	Solo founders, freelancers	\$39/mo	5 projects, 20 C-Suite runs, 30 desi
Pro	Serious founders, product manag	\$89/mo	Unlimited projects, 100 C-Suite runs
Team	Startups, agencies (3-10 seats)	\$249/mo	Everything Pro x team, shared worksp
Enterprise	Studios, VCs, accelerators	\$999+/mo	Custom seats, white-label mockups, d

Ideal Business Model

1. Primary revenue: Monthly subscription per tier
2. Expansion revenue: Extra run packs (\$9 for 25 C-Suite runs, \$9 for 50 design screens)
3. BYOK discount path: Users who connect own API keys get 20% off (saves LLM costs)
4. Annual pre-pay: 20% discount = strong cash flow + churn reduction

Unit Economics at Current LLM Costs (~\$0.012/1K tokens, Claude Sonnet)

Action	LLM Cost
C-Suite run (~150K tokens, 7 agents)	~\$0.80-1.50
Design screen generation	~\$0.15-0.40
Artifact generation	~\$0.20-0.60
Pro user at 30-40% utilization	~\$25-40/mo LLM cost
Gross margin per Pro user	~55-72% before hosting

Q4: What is the ideal monthly net revenue ballpark?

Scenario 1: Early Stage (~6 months post-launch)

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Tier	Subscribers	MRR
Indie \$39	50	\$1,950
Pro \$89	20	\$1,780
Team \$249	5	\$1,245
Enterprise \$999	1	\$999
Gross MRR		\$5,974

Cost Item	Monthly Amount
LLM (Claude Sonnet, ~30% BYOK offset)	~\$800
Fly.io sandboxes (~75 active project)	~\$600
Nhost DB + storage	~\$50
Stripe fees (2.9%)	~\$175
Inngest + misc	~\$75
Total costs	~\$1,700

Net: ~\$4,300/mo (~72% margin)

Scenario 2: Growth (~18 months)

Tier	Subscribers	MRR
Indie \$39	300	\$11,700
Pro \$89	150	\$13,350
Team \$249	40	\$9,960
Enterprise \$1,499 avg	5	\$7,495
Gross MRR		\$42,505

Cost Item	Monthly Amount
LLM (~35% BYOK, ~30% utilization)	~\$6,000
Fly.io (~500 projects x \$8)	~\$4,000
Nhost Pro	~\$200
Stripe fees	~\$1,250
Inngest + CDN + monitoring	~\$450
Total costs	~\$11,900

Net: ~\$30,600/mo (~72% margin) -> ~\$367K ARR net

Scenario 3: Scale (~3 years)

Tier	Subscribers	MRR
Indie \$39	1,000	\$39,000
Pro \$89	500	\$44,500
Team \$249	150	\$37,350
Enterprise \$1,999 avg	20	\$39,980
Annual prepay uplift	--	+\$10,000
Gross MRR		~\$170,830

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Cost Item	Monthly Amount
LLM (~40% BYOK, ~25% avg utilization)	~\$18,000
Infrastructure (Fly.io, DB, CDN, mon)	~\$18,000
Stripe + payment ops	~\$5,000
Compliance/security tools	~\$2,000
Total costs	~\$43,000

Net: ~\$127,800/mo (~75% margin) -> ~\$1.53M ARR net

Net Revenue Milestones Summary

Stage	Timeline	Subscribers	Net MRR	Net ARR
Early	~6 months	~76 paying	~\$4,300	~\$51,600
Growth	~18 months	~495 paying	~\$30,600	~\$367,200
Scale	~3 years	~1,670 paying	~\$127,800	~\$1,533,600

Key Revenue Levers

Lever	Impact
BYOK adoption hits 50%+	Saves \$3-15K/mo in LLM costs at grow
Annual prepay at 20% discount	30% annual conversion = ~20% effecti
Usage packs add-on	+\$9 packs at 15% attachment rate add
Sandbox sleep-on-idle	Active time drops 8hr -> 2-3hr/day -
Enterprise tier at \$2,500+	5 more logos = +\$12,500/mo at near-z

The \$1M net ARR milestone requires ~1,700 paying subscribers -- roughly when the async DB refactor and horizontal scaling become necessary. Conveniently, \$1M ARR also funds exactly that engineering work.